

Experiment 2

Problem Identification and Feasibility: Software Design Principles

Identify the problem addressed by the selected software project and prepare its problem statement, objectives, scope, and technical and operational feasibility analysis.

Aim

To identify the problem addressed by the AI Interview Preparation Platform and prepare its problem statement, objectives, scope, and technical and operational feasibility analysis.

Objectives

- Identify the real-world problem addressed by the software project.
- Prepare a clear problem statement.
- Define project objectives and scope.
- Perform technical and operational feasibility analysis.
- Evaluate the viability of the proposed solution.

Introduction

Problem identification is the first step in software engineering. A well-defined problem statement and feasibility study ensure that the proposed software solution is practical, achievable and aligned with stakeholder requirements before development begins.

Selected Project

Project Title: AI Interview Preparation Platform

Domain: Artificial Intelligence, Career Development and Education

Problem Statement

Traditional interview preparation requires candidates to depend on human mentors, coaching classes, or repeated practice with limited personalized feedback. Candidates often face difficulties in practicing technical and behavioral interviews, identifying weaknesses in their communication skills, and understanding whether their resume is suitable for Applicant Tracking Systems (ATS). An AI Interview Preparation Platform provides an intelligent and interactive solution by simulating technical and behavioral interviews, analyzing candidate responses, providing personalized feedback, evaluating communication and speech patterns, and analyzing resumes for ATS compatibility and skill gaps.

Project Objectives

- Provide AI-powered mock interviews for technical and behavioral interview preparation.
- Simulate realistic interview questions and interactions.
- Analyze candidate responses and provide intelligent feedback.
- Evaluate speech characteristics such as communication pacing and response quality.
- Analyze resumes for ATS compatibility.
- Identify skill gaps between the candidate's resume and desired career roles.
- Help candidates improve their interview performance and confidence.
- Provide performance metrics to track improvement over multiple practice sessions.

Project Scope

The AI Interview Preparation Platform supports:

- Technical interview simulations.
- Behavioral interview simulations.
- AI-generated or AI-selected interview questions.
- Analysis of candidate responses.
- Communication and pacing analysis.
- Response-quality evaluation.
- Personalized interview feedback.
- Resume uploading and analysis.
- ATS-compatibility evaluation.
- Skill-gap identification.
- Performance metrics and progress tracking.

Features such as actual job placement, direct recruitment, conducting real company interviews, salary negotiation, and final hiring decisions are outside the scope of this project.

Technical Feasibility Analysis

Factor	Assessment	Remarks
Technology	Feasible	Python/JavaScript, AI/ML models, REST APIs and database technologies can support development.
Hardware	Feasible	Standard computers and smartphones with microphone/camera support are sufficient.
Software Tools	Feasible	IDEs, Git, GitHub, AI/ML libraries, databases and web-development tools are available.
Security	Feasible	Authentication, authorization and secure data transmission can be implemented.

Scalability	Feasible	Cloud-based and modular architecture can support future users and features.
AI/ML	Feasible	NLP and speech-analysis techniques can be implemented for interview response evaluation.

Operational Feasibility Analysis

Factor	Assessment	Remarks
User Acceptance	High	Candidates can practice interviews through a simple AI-based interface.
Ease of Use	High	The platform provides guided interview sessions and automated feedback
Availability	High	Candidates can practice interviews anytime without depending on an interviewer.
Maintainability	High	Modular design allows AI, resume analysis and interview modules to be maintained independently.
Business Benefit	High	Reduces dependence on expensive coaching and provides personalized interview preparation.
Performance Improvement	High	Feedback and performance metrics help candidates identify and improve weaknesses.

Analysis

The feasibility study indicates that the AI Interview Preparation Platform is technically and operationally viable. Existing AI, NLP, speech-analysis, database and web-development technologies can support the implementation of the proposed system. The platform can provide candidates with accessible and personalized interview preparation without requiring a human interviewer for every practice session.

Observation

Clearly defining the problem, objectives, scope and feasibility of the AI Interview Preparation Platform provides a strong foundation for its subsequent software design, development and implementation.

Result

The problem statement, objectives, project scope, technical feasibility analysis and operational feasibility analysis for the AI Interview Preparation Platform were successfully prepared.